REFLOWECONOMY · PROJECT PLAN

ReflowEconomy project plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First project plan (kit 1.8 pipeline) at stage P3, with the reference list of every source cited in the repo

ReflowEconomy project plan

ReflowEconomy is at stage P3 (design outputs) and TRL 3, the portfolio cap. The next gate, P4 (verification, TRL 4), is on hold by Amish's instruction; before it, three decisions are requested: how the site meets its supply limit (RFE-DEC-001 D1), which concept is the reference (D4), and the approach to the first co-design partner. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Playbook content	Design lead	Feasibility matrix with hazard, export point and sources (PR-001); export rules per stream (PR-004); food-contact rule from IS 14534	Open: PR-001 and PR-004 not met
Reference micro-factory design	Design lead	Model at D3, drawings, priced BOM, build plan, calculation note	Done at P3; supply decision open (D1)
Concept selection	Amish	Choice between A, B and the staged start (RFE-OPT-001, RFE-DDR-004)	Proposed, awaiting Amish (D4)
Material passport	Design lead	Schema v0.2 with examples; a phone form for PR-015	Schema done; form not started (P4)
Co-design	Amish with the partner	Sessions with a waste-picker cooperative (first candidate SWaCH through WIEGO)	Not started; nothing agreed
Regulatory	Operator with a consultant	Registration and consent route in Maharashtra (RFE-REG-001)	To assess
Verification (P4)	Design lead with the partner	Test plans and reports for the first checks in RFE-BLD-001 section 5	On hold (TRL cap 3)

Resources needed

Table 2. Resources (indicative; for planning, not quotes).

Resource	Deliverable	Effort	Estimated cost (USD)
Co-design facilitation with a waste-picker cooperative	Validated user requirements, input mix, prices, wages, working day	3 sessions	To assess
Licensed electrician (partner region)	Supply design for the option chosen in D1; installation certificate at P4	1 to 2 days of design	To assess
Qualified safety professional	Review of layout, exits, headroom, guarding and interlocks	2 days	To assess
Structural engineer (container option only)	Seam beam and propping design	1 to 2 days	To assess
Equipment for a first site (P4, on hold)	The reference micro-factory, `bom/bom.csv` items 1 to 12	Build to RFE-BLD-001	25,505 indicative (value-engineering target 25,000)

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Done 2026-10-25 (kit 1.9.1 upgrade)
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT	Done 2026-10-25
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed, feasibility closed	Done except the STEM check (open on S14, S15, S21) and the concept choice (awaiting Amish)
P3 Design outputs	Model at D3, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability complete	Reached 2026-10-25
P4 Verification	Test plans and lab reports for the first checks; air, noise and current measurements	On hold (TRL cap 3)
P5 Validation	Field trial with the partner; registration and consents	On hold
P6 Commercialization	Launch plan, operator training, standard cost; only with a named deployment partner	On hold

Next steps and decision requested

1. Amish to decide D1 (supply: three-phase recommended, single-phase in three blocks as the fallback) and D4 (concept: the staged start recommended).
2. Approach SWaCH through WIEGO for co-design, as decided on 2026-10-02.
3. Write the missing playbook columns and export rules (PR-001, PR-004).
4. TRL 4 stays on hold until Amish lifts the cap.

References

Every source cited in this repo's documents, Chicago style (all checked online on the dates given in the citing documents).

- ABB. *Technical Note: IEC 60034-30-1 Standard on Efficiency Classes for Low Voltage AC Motors*. 2018. https://library.e.abb.com/public/db64d153e3c346938e18916e66fb1d0d/9AKK107319%20EN%2005-2018_20848_ABB_Technical_note_IEC_60034_30_1.pdf.
- Axelent. "Standards." https://www.axelent.dk/media/3343/normsen.pdf.
- Basel Convention. "Plastic Waste Amendments." https://www.basel.int/implementation/plasticwaste/amendments/overview/tabid/8426/default.aspx.
- Borealis. *Statement on Impurities: Polyolefins*. Edition 6, 2022. https://www.borealisgroup.com/storage/Emission-EN-ed.06.pdf.
- Brooks, Amy L., Shunli Wang, and Jenna R. Jambeck. "The Chinese Import Ban and Its Impact on Global Plastic Waste Trade." *Science Advances* 4, no. 6 (2018). https://doi.org/10.1126/sciadv.aat0131.
- Bureau of Indian Standards. *IS 14534:1998, Guidelines for Recycling of Plastics*. https://law.resource.org/pub/in/bis/S11/is.14534.1998.pdf.
- Elliott Electric Supply. "Full Load Current in Amps for Single Phase AC Motors" (NEC Table 430.248). https://elliottelectric.com/StaticPages/ElectricalReferences/ElectricalTables/Amperes_Single_Phase_Alternate.aspx.
- European Union. Regulation (EU) 2023/1542 on batteries. https://eur-lex.europa.eu/eli/reg/2023/1542/oj.
- European Union. Regulation (EU) 2024/1781, Ecodesign for Sustainable Products. https://eur-lex.europa.eu/eli/reg/2024/1781/oj.
- factoHR. "Occupational Safety, Health and Working Conditions (OSH) Code, 2020." https://factohr.com/occupational-safety-health-and-working-conditions-code/.
- Government of India. *The Factories Act, 1948*. https://rskr.irimee.in/sites/default/files/FACTORIESACT-CPN.pdf.
- Government of India, MoEFCC. *Solid Waste Management Rules, 2016*. https://cdnbbsr.s3waas.gov.in/s30f46c64b74a6c964c674853a89796c8e/uploads/2024/07/20240710555191345.pdf.
- HDFC Sky. "Rupee Rises 24 Paise to Close at 95.46 Against US Dollar." August 25, 2026. https://hdfcsky.com/news/rupee-rises-24-paise-to-close-at-95-46-against-us-dollar.
- IndiaMART. "PET Bottle Flake and Crush"; "PET Flakes Scrap"; "Recycled Plastic Sheets, Panels and Boards"; S M Engineering, Chennai, "10 Ton Compact Baling Press." https://m.indiamart.com/impcat/pet-bottle-flake.html; https://m.indiamart.com/impcat/pet-flakes-scrap.html; https://m.indiamart.com/impcat/recycled-plastic-sheets.html; https://m.indiamart.com/smengineeringchennai.
- International Aluminium Institute. "Aluminium Recycling Saves 95 % of the Energy Needed for Primary Aluminium Production." https://international-aluminium.org/landing/aluminium-recycling-saves-95-of-the-energy-needed-for-primary-aluminium-production/.
- ITU and UNITAR. *Global E-waste Monitor 2024*. https://ewastemonitor.info/the-global-e-waste-monitor-2024/.
- Maharashtra Pollution Control Board. "Registration of Plastic Waste Processor." https://mpcb.gov.in/sites/default/files/plastic_processor.pdf.
- Maharashtra State Electricity Distribution Co. *Conditions of Supply*. 2010. https://www.mahadiscom.in/wp-content/uploads/2018/08/CoS_MSEDCL_2010.pdf.
- OECD. *Global Plastics Outlook*. 2022. https://www.oecd.org/en/publications/global-plastics-outlook_de747aef-en.html.
- OSHA. "Table Z-1, Limits for Air Contaminants." https://www.osha.gov/annotated-pels/table-z-1.

- Plastics Technology. "How to Dry PET for Container Applications." https://www.ptonline.com/articles/how-to-dry-pet-for-container-applications.
- Precious Plastic. Academy pages for the Shredder Pro, Extrusion Pro and sheetpress; Bazar listings for the V4 extrusion, the 1 x 1 m hot sheet press, the sheet press workshop and the Shredder Pro 3.0. https://www.preciousplastic.com; https://bazar.preciousplastic.com.
- RJC Mold. "Plastic Melting Point Chart." https://rjcmold.com/guides/plastic-melting-point-chart.
- Schneider Electric. ATV12HU22M2 and ATV12HU15M2 product data sheets. https://media.cetnaj.com.au/Datasheets/Schneider/Schneider_ATV12HU22M2.pdf; https://www.mouser.com/datasheet/2/357/Altivar_12_ATV12HU15M2-1782510.pdf.
- Szarka, Györgyi, and Béla Iván. "Thermal Properties, Degradation and Stability of Poly(vinyl chloride) Predegraded Thermooxidatively in the Presence of Dioctyl Phthalate Plasticizer." *Journal of Macromolecular Science, Part A* 50, no. 2 (2013): 208 to 214. https://real.mtak.hu/4463/.
- TIS (GDV). "Permissible Loading Capacity of Containers." https://www.tis-gdv.de/tis_e/containe/belast/belast-htm/.
- Troax. *Safety Guide*. https://www.troax.com/app/uploads/Brochure-Safety-Guide_EN.pdf.
- UNEP. *Global Waste Management Outlook 2024*. https://www.unep.org/resources/global-waste-management-outlook-2024.
- US EPA. *An Analysis of Lithium-ion Battery Fires in Waste Management and Recycling*. 2021. https://www.epa.gov/system/files/documents/2021-08/lithium-ion-battery-report-update-7.01_508.pdf.
- WHO. *Children and Digital Dumpsites*. 2021. https://www.who.int/publications/i/item/9789240023901.
- WIEGO. "Fact Sheet: Cooperative SWaCH." https://wiego.org/wp-content/uploads/2019/09/fact_sheet_cooperative_swach.pdf.
- World Bank. *What a Waste 2.0*. 2018. https://datatopics.worldbank.org/what-a-waste/.
- Standards cited without a public text: ISO 13857:2019, ISO 13849-1, ISO 14119, ISO 14021, ISO 11228-1, IEC 60204-1, IEC 60034-30-1, NFPA 92, ACGIH *Industrial Ventilation*, IS 875 (Part 2), National Building Code of India 2016, EU Directive 2003/10/EC; B. Wunderlich, *Thermal Analysis of Polymeric Materials* (Springer, 2005).